Synthetic Falsifiers for Recognition Science Predictions in Genome Architecture

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Abstract

We define a formal framework for testing the Recognition Science (RS) theory of genome architecture against synthetic biology constructs. Three machine-checked assay predicates are formalized in Lean 4: (1) a **degeneracy assay** that tests whether a synthetic code preserves the standard degeneracy pattern; (2) a **wobble assay** that tests whether the clean-wobble family-box count is maintained; and (3) an **optimality assay** that tests whether the synthetic code achieves lower total weighted mutation cost than the standard code. A synthetic construct that fails any assay constitutes a formal falsifier of the RS framework. The standard code is proved to pass all three assays. We describe five classes of synthetic constructs—reduced-redundancy codes, phase-misaligned introns, sub-threshold introns, repeat-depleted genomes, and clock-disrupted timing elements—each designed to probe a specific layer of the RS five-layer genome decomposition. This is a design paper: it specifies the constructs, assays, and predicted outcomes, but does not report experimental results. The purpose is to convert the RS framework from a post-hoc interpretation into an experimentally falsifiable program.

1 Introduction

A theoretical framework that cannot be falsified is not science. The RS theory of genome architecture [8, 4] makes specific structural predictions: the standard genetic code minimizes a position-weighted mutation cost [2], introns are positioned by helical phase stress [6], wobble degeneracy dominates the codon table [3], and the FOLD operator confers a categorical advantage to sexual reproduction [5].

Each of these predictions can in principle be tested by constructing synthetic codes or genomic architectures that violate the predicted constraints and measuring the phenotypic consequences. This paper defines the formal assay framework and specifies five classes of synthetic falsifier constructs.

The key contribution is that the assay predicates are **formalized in Lean 4**. The wobble assay is decidable (equality of natural numbers); the degeneracy and optimality assays are logical propositions over the formalized code. The standard code is proved to pass all three assays, so any construct that fails at least one is a formal falsifier—a machine-checkable witness that the RS prediction is violated.

2 The Assay Framework

2.1 Synthetic Constructs

A **synthetic construct** is a codon assignment—a bijection on the 20 amino acid labels that relabels which amino acid each codon family encodes. The standard code is the identity assignment:

```
standardConstruct has assignment :=
standardAssignment
```

(equivalently, `ofAssignment standardAssignment`). Both are defined in `SyntheticFalsifierAssays.lean`.

2.2 Three Assay Predicates

Assay 1: Degeneracy. A construct passes the degeneracy assay if and only if its assignment respects

the standard degeneracy classes:

$$\texttt{degeneracyAssay}(s) \Leftrightarrow \texttt{respectsDegeneracyClass}(s.\texttt{assignment}).$$

A construct that reassigns a 6-fold amino acid to a 2-fold codon family fails this assay.

Assay 2: Wobble. A construct passes the wobble assay if the number of family boxes with clean wobble symmetry (all four third-position variants synonymous, or a clean pyrimidine/purine 2+2 split) equals the standard count:

$$\texttt{wobbleAssay}(s) \Leftrightarrow \texttt{assignedCleanWobbleCount}(s.\texttt{assignment}) = \texttt{cleanWobbleCount}.$$

The standard code has 14 of 16 clean family boxes [3]. A reassignment that breaks this count fails the wobble assay.

Assay 3: Optimality. A construct passes the optimality assay if the standard code's total weighted mutation cost is no greater than the construct's cost:

$$\texttt{optimalityAssay}(w, s) \Leftrightarrow \texttt{standardAssignmentCost}(w) \leq \texttt{totalAssignmentCost}(w, s.\texttt{assignment}).$$

A construct that achieves strictly lower cost than the standard code under the chosen weight vector w fails the optimality assay.

2.3 Falsifier Definition

A construct is **RS-violating** if it fails at least one of the three assays:

$$\texttt{rsViolatingConstruct}(w, s) \Leftrightarrow \neg\,\texttt{degeneracyAssay}(s) \vee \neg\,\texttt{wobbleAssay}(s) \vee \neg\,\texttt{optimalityAssay}(w, s).$$

A `SyntheticFalsifier` packages the construct together with a proof that it violates at least one assay.

2.4 The Standard Code Passes All Assays

> **Theorem** (`standardConstruct_not_falsifier`). For any weight vector w, the standard code is not an RS-violating construct: $\neg\,\texttt{rsViolatingConstruct}(w, \texttt{standardConstruct})$.

The proof proceeds by showing that the standard code passes each assay individually:

- `standardConstruct_passes_degeneracyAssay`: the identity assignment respects all degeneracy classes.
- `standardConstruct_passes_wobbleAssay`: the identity assignment preserves the clean-wobble count.
- `standardConstruct_passes_optimalityAssay`: the identity assignment's cost equals its own cost ($\leq$ is reflexive).

3 Five Classes of Falsifier Constructs

Each class targets a specific layer of the RS five-layer genome decomposition [4].

3.1 Class 1: Reduced-Redundancy Codes

Target layer: Error correction.

Construct: Reassign amino acids to reduce average degeneracy below the standard $61/20 \approx 3.05$. For example, split current 4-fold families into pairs of 2-fold families by introducing new amino acid assignments at the wobble position.

Predicted assay outcome: Fails the wobble assay (clean-wobble count drops below 14) and may fail the optimality assay (reduced redundancy increases vulnerability to point mutations).

Falsifier for RS: If a reduced-redundancy code shows *improved* fitness in a controlled expression system, the RS prediction that the current redundancy level is error-optimal is wrong.

3.2 Class 2: Phase-Misaligned Introns

Target layer: Phase alignment.

Construct: Insert synthetic introns at positions that violate the helical phase-stress model: not on the half-octave grid ($p \not\equiv 0 \pmod 4$), and at low-stress rather than high-stress sites.

Predicted assay outcome: Reduced splicing efficiency or gene expression, measurable by RT-qPCR or ribosome profiling.

Falsifier for RS: If phase-misaligned introns produce *equal or better* expression than phase-aligned introns of the same length, the helical-phase model [6] is wrong.

3.3 Class 3: Sub-Threshold Introns

Target layer: Phase alignment (minimal intron size).

Construct: Engineer introns shorter than the predicted minimal size of $8\varphi^5 \approx 89$ bp. Test introns of 50, 60, 70, and 80 bp.

Predicted assay outcome: Introns below ∼85 bp should show impaired splicing or reduced expression in mammalian cells.

Falsifier for RS: If introns of 50–60 bp splice efficiently and support normal expression in mammalian cells, the $8\varphi^5$ prediction [6] is wrong.

3.4 Class 4: Repeat-Depleted Genomes

Target layer: Ledger balance.

Construct: Systematically delete repetitive elements (SINEs, LINEs, LTR retrotransposons) from a defined genomic region while preserving all annotated genes and regulatory elements.

Predicted assay outcome: Impaired gene regulation, chromatin organization, or replication timing in the depleted region.

Falsifier for RS: If repeat depletion has no measurable phenotypic consequence (beyond the directly deleted elements), the ledger-balance hypothesis [4] is wrong.

3.5 Class 5: Clock-Disrupted Timing Elements

Target layer: Temporal coordination.

Construct: Mutate or delete non-coding elements predicted to implement φ-ladder timing (e.g., clock gene regulatory regions, replication-timing control elements) and measure the effect on downstream timing phenotypes.

Predicted assay outcome: Disrupted biological timing at the specific φ-ladder rung corresponding to the deleted element.

Falsifier for RS: If deletion of a predicted clock element has no effect on any timing phenotype, the gearbox model [7] is wrong for that rung.

4 Formal Verification

The assay predicates and all structural theorems are formalized in Lean 4 (version 4.27.0) with Mathlib. The principal module is `SyntheticFalsifierAssays.lean`, which imports:

- `CodonAssignmentGlobalOptimality.lean`: the `GlobalAssignmentCounterexampleCertificate` type.
- `CodonAssignmentOptimality.lean`: the `swapImproves` predicate.
- `FamilyBoxAnalysis.lean`: `allFamilyBoxes`, `cleanWobbleCount`, `hasCleanWobble`.
- `DegeneracySearchSpace.lean`: `respectsDegeneracyClass`, `standardAssignment_respectsDegeneracyClass`.

Key theorems:

- `standardConstruct_not_falsifier`: the standard code passes all three assays.
- `degeneracyFailure_yields_falsifier`: any assignment that violates degeneracy classes is a falsifier.
- `wobbleFailure_yields_falsifier`: any assignment that changes the clean-wobble count is a falsifier.
- `costImprovement_yields_falsifier`: any assignment that strictly reduces total cost is a falsifier.
- `improvingSwap_yields_falsifier`: any improving pairwise swap is a falsifier.
- `globalCounterexample_yields_falsifier`: any global counterexample certificate is a falsifier.

The module contains zero axioms and zero `sorry` assertions.

5 Discussion

5.1 Why a Design Paper

This paper specifies constructs, assays, and predictions but does not report experimental results. The purpose is to convert the RS framework from a post-hoc interpretation into a pre-registered experimental program. Each construct class can be independently tested by a synthetic biology or genomics laboratory without requiring acceptance of the broader RS theory.

5.2 Prioritization

The five construct classes differ in experimental difficulty:

Class	Difficulty	Technology
1. Reduced redundancy	Medium	Expanded genetic codes
2. Phase-misaligned introns	Low	Synthetic gene design
3. Sub-threshold introns	Low	Minigene reporters
4. Repeat depletion	High	CRISPR-based deletion
5. Clock disruption	High	Enhancer/promoter KO

Classes 2 and 3 are the easiest to test and should be prioritized. A single well-controlled experiment showing that phase-misaligned introns splice as well as phase-aligned ones would refute the helical-phase model.

5.3 Relation to Existing Synthetic Code Work

The expanded genetic code community [1] has already engineered organisms with reassigned codons (e.g., *E. coli* with all TAG codons reassigned). These experiments test codon *identity*, not codon *optimality*. The RS falsifier framework asks a different question: not whether a reassigned code *can work*, but whether it *outperforms* the standard code on the RS-predicted objective.

5.4 Scope

The assay framework currently covers codon-level properties (degeneracy, wobble, cost). It does not yet cover intron-level or timing-level properties, which would require extending the Lean formalization with sequence-level predicates. Classes 2–5 are specified as construct descriptions, not as formalized assay predicates.

6 Conclusion

The RS theory of genome architecture generates specific, testable predictions at every layer of the five-layer decomposition. This paper defines a formal assay framework (three machine-checked predicates) and five classes of synthetic falsifier constructs designed to probe each layer independently. The standard code is proved to pass all assays. Any construct that fails at least one assay is a formal, machine-checkable falsifier. The immediate experimental priorities are phase-misaligned introns and sub-threshold introns, both of which can be tested with standard synthetic gene design.

Data Availability

All Lean source files are in the `reality` repository under `IndisputableMonolith/Genetics/`.

References

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